

Enterprise Linux Hardening & Defensive Security

Internship Project Final Presentation

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PROJECT OBJECTIVES

Security Goal

Attack Surface Reduction

Systematically disabling unused services and securing entry points to minimize exploit vectors.

Compliance

Industry Standards

Aligning system configurations with CIS (Center for Internet Security) Benchmarks.

TECHNICAL SECURITY STACK

Tool	Category	Primary Function
Nessus Professional	Vulnerability Mgmt	Automated network scanning & risk assessment.
Fail2Ban	Intrusion Prevention	Log-based blocking of brute-force attack patterns.
UFW Firewall	Network Security	Stateful packet filtering & port management.
Auditd	Kernel Auditing	System-level logging of user & file activities.
Lynis	Security Auditing	Compliance scoring and hardening suggestions.

PHASE 1: PERIMETER HARDENING

In the initial phase, the focus was on securing the "front door" of the server.

- **SSH Protocol Hardening:** Disabled password-based authentication; enforced ED25519 Elliptic Curve keys.
- **Port Sanitization:** Reconfigured UFW with a "Default Deny" policy for both ingress and egress traffic.
- **Service Masking:** Identified and masked 12+ non-essential systemd services to reduce memory footprint and risk.

PHASE 2: ACTIVE DEFENSE & MONITORING

Implementation of reactive security measures and deep-system visibility.

Fail2Ban Configuration

Created custom jail definitions for SSH. Successfully banned 45+ unique malicious IP addresses within the first 24 hours of monitoring.

Auditd Rule Implementation

Configured watches on `/etc/shadow` and `/bin/bash` to detect unauthorized privilege escalation attempts.

PHASE 3: VULNERABILITY ANALYSIS

Comprehensive scanning using **Nessus Professional**.

Initial Findings

- 3 High Severity Vulnerabilities (Kernel Exploit risk).
- Weak SSH Cipher support (CBC modes).
- ICMP Timestamp responses enabled.

Remediation Path

- Automated Patch Orchestration via APT.
- SSH Configuration tuned for NIST compliance.
- Kernel parameter tuning via sysctl.

QUANTITATIVE PROJECT IMPACT

100%

Critical Patches Applied

+32 pts

Lynis Hardening Score

0%

Post-Fix Breach Attempts

CONCLUSION & FUTURE SCOPE

- **Outcome:** Successfully delivered a production-ready, verified secure Linux environment.
- **Key Learning:** Mastered the intersection of system administration and defensive security operations.
- **Recommendation:** Automate these hardening steps using Ansible playbooks for cross-departmental scalability.

Thank You

Questions?